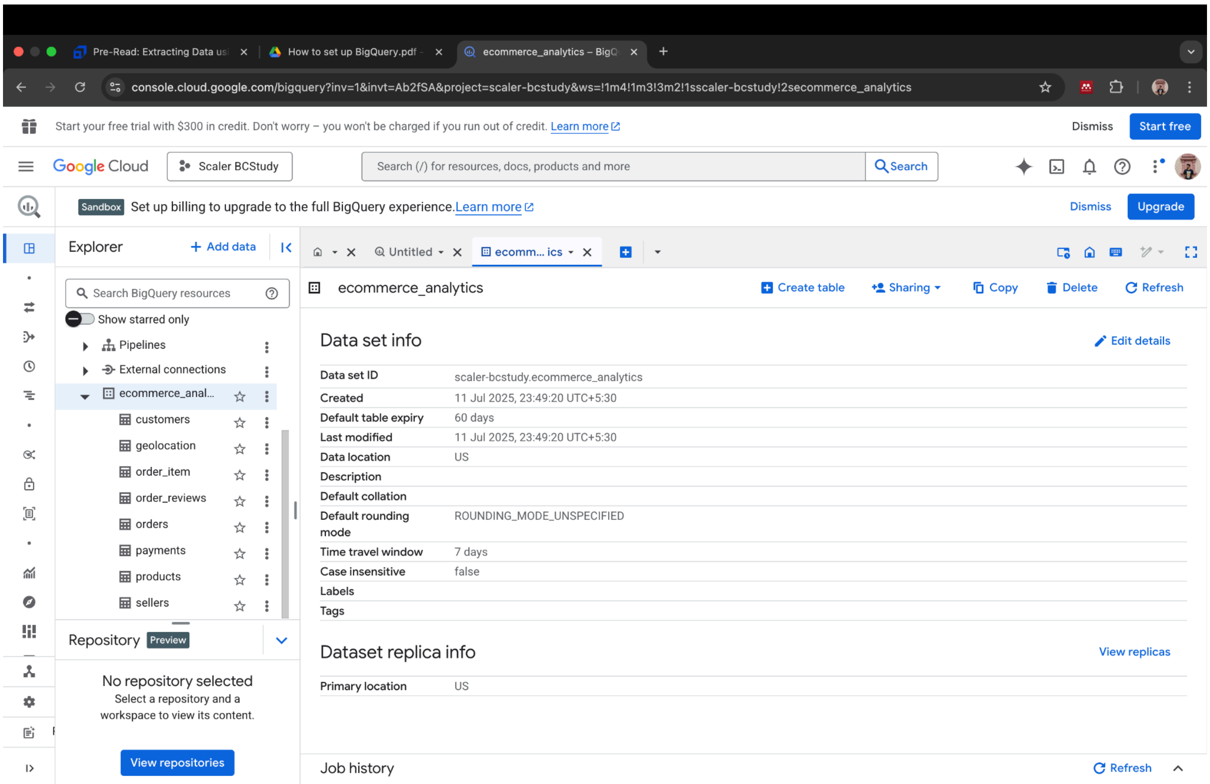
Step 1: Set Up Your Project in Google BigQuery



Step 2: Run Inspection Queries in BigQuery

1. Customers table:

SQL:

```
SELECT * FROM `scaler-bcstudy.ecommerce_analytics.customers` LIMIT 5;
```

RESULTS:

customer_id	customer_unique_id	customer_zip_code_prefix	customer_city	customer_state
2201362e68992f654942dc0067c1b716	f7d7fc0a59ef4363fdce6e3aa069d498	69900	rio branco	AC
31dbc13addc753e210692eacaea065e4	5dbba6c01268a8ad43f79157bf4454a0	69900	rio branco	AC
dad907e170748a35ef4e92238b7308f3	36b1c0516f123351ffa87430416dcae5	69900	rio branco	AC
888d2ebe1af2a8c93c75dae5dfc23719	721d1092e1a6460c67e6a0e691d899a3	69900	rio branco	AC
8a0108267d9258a0ec9f74381bc9b0de	7a2dc4682890550ebe3b8befcea3d55c	69900	rio branco	AC

2. Geolocation Table:

SQL:

```
SELECT * FROM `scaler-bcstudy.ecommerce_analytics.geolocation` LIMIT 5;
```

RESULTS:

geolocation_zip_code_prefix	geolocation_lat	geolocation_lng	geolocation_city	geolocation_state
1037	-23.54562128115270	-46.639292048001700	sao paulo	SP
1046	-23.546081127035500	-46.64482029837160	sao paulo	SP
1046	-23.546128966414700	-46.642951483611400	sao paulo	SP
1041	-23.5443921648681	-46.639499306278400	sao paulo	SP
1035	-23.541577961711500	-46.641607223296100	sao paulo	SP

3. Order_item Table:

SQL:

SELECT * FROM `scaler-bcstudy.ecommerce_analytics.order_item` LIMIT 5;

RESULTS:

order_id	order_item_id	product_id	seller_id	shipping_limit_date	price	freight_value
3ee6513ae7ea23bdfab5b9ab60bffcb5	1	8a3254bee785a526d548a81a9bc3c9be	96804ea39d96eb908e7c3afdb671bb9e	2018-05-04 03:55:26.000000 UTC	0.85	18.23
6e864b3f0ec71031117ad4cf46b7f2a1	1	8a3254bee785a526d548a81a9bc3c9be	96804ea39d96eb908e7c3afdb671bb9e	2018-05-02 20:30:34.000000 UTC	0.85	18.23
c5bdd8ef3c0ec420232e668302179113	2	8a3254bee785a526d548a81a9bc3c9be	96804ea39d96eb908e7c3afdb671bb9e	2018-05-07 02:55:22.000000 UTC	0.85	22.3
8272b63d03f5f79c56e9e4120aec44ef	2	05b515fdc76e888aada3c6d66c201dff	2709af9587499e95e803a6498a5a56e9	2017-07-21 18:25:23.000000 UTC	1.2	7.89
8272b63d03f5f79c56e9e4120aec44ef	3	05b515fdc76e888aada3c6d66c201dff	2709af9587499e95e803a6498a5a56e9	2017-07-21 18:25:23.000000 UTC	1.2	7.89

4. Order_reviews Table:

SQL:

SELECT * FROM `scaler-bcstudy.ecommerce_analytics.order_reviews` LIMIT 5;

RESULTS:

review_id	order_id	review_score	review_comment_title	review_creation_date	review_answer_timestamp
9e628acdf082a5eeea001b1ded5d94a1	54d744a4410b1edccc36c6d1f1c7e097	1		0001-02-17 00:00:00.000000 UTC	0015-03-17 02:47:00.000000 UTC
db27636ea82d449a9e0d1bea6b419465	a9a93c428c6103f2151bb63a1d32a520	1		0001-02-17 00:00:00.000000 UTC	0001-02-17 11:12:00.000000 UTC
5bec0620e043423a063641e9a2900028	1cb796218c383fc54a6a4541420201d8	1		0001-02-17 00:00:00.000000 UTC	0013-02-17 11:43:00.000000 UTC
3c9d8bbd2a92baddc2356f4c8ee4563c	b3feb3846bb0a8d68cd328138324bea4	1		0001-02-18 00:00:00.000000 UTC	0002-02-18 10:36:00.000000 UTC

647a5c7d693aedb5ce12f3c5e5ec9095	745e2506fb647deca4669e1c88ce9783	1		0001-02-18 00:00:00.000000 UTC	0001-02-18 21:10:00.000000 UTC

5. Orders Table:

SQL:

SELECT * FROM `scaler-bcstudy.ecommerce_analytics.orders` LIMIT 5;

RESULTS:

order_id	customer_id	order_status	order_purchase_timestamp	order_approved_at	order_delivered_carrier_date	order_delivered_customer_date	order_estimated_delivery_date
a2e4c44360b4a57bdff22f3a4630c173	8886130db0ea6e9e70ba0b03d7c0d286	approved	2017-02-06 20:18:17.000000 UTC	2017-02-06 20:30:19.000000 UTC			2017-03-01 00:00:00.000000 UTC
132f1e724165a07f6362532bfb97486e	b2191912d8ad6eac2e4dc3b6e1459515	approved	2017-04-25 01:25:34.000000 UTC	2017-04-30 20:32:41.000000 UTC			2017-05-22 00:00:00.000000 UTC
809a282bbd5dbcabb6f2f724fca862ec	622e13439d6b5a0b486c435618b2679e	canceled	2016-09-13 15:24:19.000000 UTC	2016-10-07 13:16:46.000000 UTC			2016-09-30 00:00:00.000000 UTC
e5215415bb6f76fe3b7cb68103a0d1c0	b6f6cbfc126f1ae6723fe2f9b3751208	canceled	2016-10-22 08:25:27.000000 UTC				2016-10-24 00:00:00.000000 UTC
71303d7e93b399f5bcd537d124c0bcfa	b106b360fe2ef8849fbbd056f777b4d5	canceled	2016-10-02 22:07:52.000000 UTC	2016-10-06 15:50:56.000000 UTC			2016-10-25 00:00:00.000000 UTC

6. Payments Table:

SQL:

SELECT * FROM `scaler-bcstudy.ecommerce_analytics.payments` LIMIT 5;

RESULTS:

order_id	payment_sequential	payment_type	payment_installments	payment_value
744bade1fcf9ff3f31d860ace076d422	2	credit_card	0	58.69
1a57108394169c0b47d8f876acc9ba2d	2	credit_card	0	129.94
8bcbe01d44d147f901cd3192671144db	4	voucher	1	0.0
fa65dad1b0e818e3ccc5cb0e39231352	14	voucher	1	0.0
6ccb433e00daae1283ccc956189c82ae	4	voucher	1	0.0

7. Products Table:

SQL:

SELECT * FROM `scaler-bcstudy.ecommerce_analytics.products` LIMIT 5;

RESULTS:

product_id	product category	product_name_length	product_description_length	product_photos_qty	product_weight_g	product_length_cm	product_height_cm	product_width_cm
5eb564652db742ff8f28759cd8d2652a								
09ff539a621711667c43eba6a3bd8466	babies	60	865	3				
a69f15dfb803d485e8933e80b9742c1c	Watches present	53	506	6	150	11	16	6
2f763ba79d9cd987b2034aac7ceffe06	electronics	45	1198	2	595	8	6	6
106392145fca363410d287a815be6de4	bed table bath	58	309	1	2083	12	2	7

8. **Sellers Table:**

SQL:

SELECT * FROM `scaler-bcstudy.ecommerce_analytics.sellers` LIMIT 5;

RESULTS:

seller_id	seller_zip_code_prefix	seller_city	seller_state
c13ef0cfbe42f190780f621ce81f2234	1207	sao paulo sp	SP
5444b12c82f21c923f2639ebc722c1ea	2051	sao pauo	SP
1cbd32d00d01bb8087a5eb088612fd9c	3363	sp / sp	SP
71593c7413973a1e160057b80d4958f6	3407	sao paulo / sao paulo	SP
6f1a1263039c76e68f40a8e536b1da6a	3581	sao paulop	SP

Step 3: Understand Relationships Between Tables

Image

Step 4: Example Use Cases You Can Query

1. Total number of orders

SQL:

```
SELECT COUNT(*) AS total_orders
FROM `scaler-bcstudy.ecommerce_analytics.orders`;
```

RESULTS:

total_orders
99441

2. Top 10 most sold products

SQL:

```
SELECT
  product_id,
  COUNT(*) AS order_count
FROM `scaler-bcstudy.ecommerce_analytics.order_item`
GROUP BY product_id
ORDER BY order_count DESC
LIMIT 10;
```

RESULTS:

product_id	order_count
aca2eb7d00ea1a7b8ebd4e68314663af	527
99a4788cb24856965c36a24e339b6058	488
422879e10f46682990de24d770e7f83d	484
389d119b48cf3043d311335e499d9c6b	392
368c6c730842d78016ad823897a372db	388
53759a2ecddad2bb87a079a1f1519f73	373
d1c427060a0f73f6b889a5c7c61f2ac4	343
53b36df67ebb7c41585e8d54d6772e08	323
154e7e31ebfa092203795c972e5804a6	281
3dd2a17168ec895c781a9191c1e95ad7	274

3. Total revenue by payment type

SQL:

```
SELECT
  payment_type,
  ROUND(SUM(payment_value), 2) AS total_revenue
FROM `scaler-bcstudy.ecommerce_analytics.payments`
GROUP BY payment_type
ORDER BY total_revenue DESC;
```

RESULTS:

payment_type	total_revenue
credit_card	12542084.19

UPI	2869361.27
voucher	379436.87
debit_card	217989.79
not_defined	0.0

4. Best reviewed products (average score > 4.5)

SQL:

```
SELECT
  oi.product_id,
  AVG(r.review_score) AS avg_score,
  COUNT(*) AS review_count
FROM `scaler-bcstudy.ecommerce_analytics.order_reviews` r
JOIN `scaler-bcstudy.ecommerce_analytics.orders` o
  ON r.order_id = o.order_id
JOIN `scaler-bcstudy.ecommerce_analytics.order_item` oi
  ON o.order_id = oi.order_id
GROUP BY oi.product_id
HAVING avg_score > 4.5
ORDER BY avg_score DESC
LIMIT 10;
```

RESULTS:

product_id	avg_score	review_count
dca00aeaa69da64cdecdb72e1c205111	5.0	1
e07fbd0969dcddca941b8c4adec498b1	5.0	3
c69da983317ecee971f64fc8202c959a	5.0	5
8bb6ace4cf2411747ea1cfecad5a458d	5.0	1
8481b767d90ff5c67263520bf912b6c4	5.0	1
d696ead711c63f9d5d1af6fcfbc3fca2	5.0	1
f3b82a9bd4330abce1d6fc15a966e6d7	5.0	5
715c4b03fc180db258f11f6808749674	5.0	2
5bfcd193eb29c1c8332e2d4ca804d54f	5.0	5
1d828db422e61ea21702baf9eda8732d	5.0	1

5. Segment by State and Total Orders

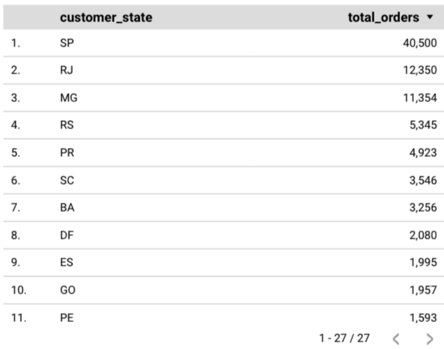
SQL:

```
SELECT
  c.customer_state,
  COUNT(DISTINCT o.order_id) AS total_orders,
```

```
  ROUND(SUM(p.payment_value), 2) AS total_payment
FROM `scaler-bcstudy.ecommerce_analytics.orders` o
JOIN `scaler-bcstudy.ecommerce_analytics.customers` c
  ON o.customer_id = c.customer_id
JOIN `scaler-bcstudy.ecommerce_analytics.payments` p
  ON o.order_id = p.order_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY total_orders DESC;
```

RESULTS:

orders



	customer_state	total_orders ▾
1.	SP	40,500
2.	RJ	12,350
3.	MG	11,354
4.	RS	5,345
5.	PR	4,923
6.	SC	3,546
7.	BA	3,256
8.	DF	2,080
9.	ES	1,995
10.	GO	1,957
11.	PE	1,593

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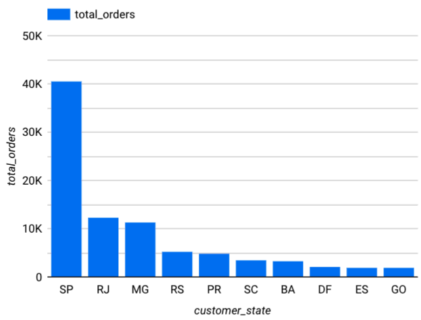
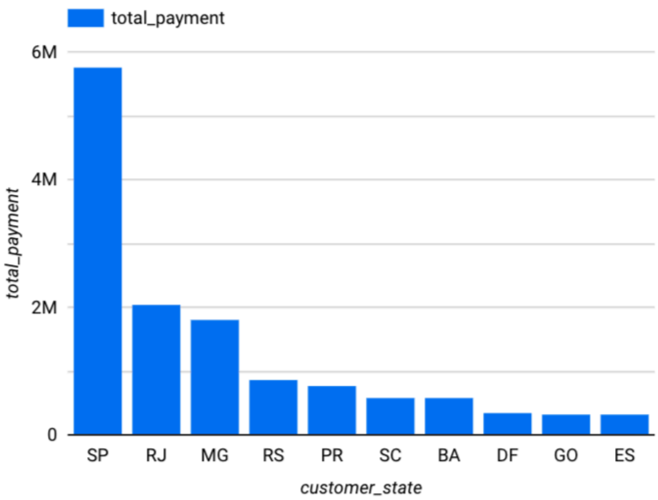


6. Total Payment Value per State



SQL:

```
SELECT
 c.customer_state,
 COUNT(DISTINCT o.order_id) AS total_orders,
 ROUND(SUM(p.payment_value), 2) AS total_payment
FROM `scaler-bcstudy.ecommerce_analytics.orders` o
JOIN `scaler-bcstudy.ecommerce_analytics.customers` c
 ON o.customer_id = c.customer_id
JOIN `scaler-bcstudy.ecommerce_analytics.payments` p
 ON o.order_id = p.order_id
WHERE o.order_status = 'delivered'
GROUP BY c.customer_state
ORDER BY total_payment DESC;
```

7. Segment Customers by Spend

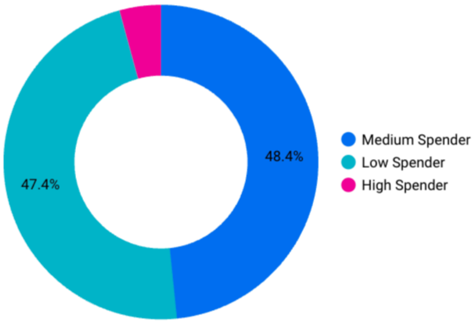
SQL:

```
WITH customer_spend AS (
  SELECT
    c.customer_unique_id,
    ROUND(SUM(p.payment_value), 2) AS total_spent,
    COUNT(DISTINCT o.order_id) AS total_orders,
    ROUND(SUM(p.payment_value) / COUNT(DISTINCT o.order_id), 2) AS
avg_spend
  FROM `scaler-bcstudy.ecommerce_analytics.orders` o
  JOIN `scaler-bcstudy.ecommerce_analytics.customers` c
    ON o.customer_id = c.customer_id
  JOIN `scaler-bcstudy.ecommerce_analytics.payments` p
    ON o.order_id = p.order_id
  WHERE o.order_status = 'delivered'
  GROUP BY c.customer_unique_id
),
segmented AS (
  SELECT *,
    CASE
      WHEN avg_spend <= 100 THEN 'Low Spender'
      WHEN avg_spend <= 500 THEN 'Medium Spender'
      ELSE 'High Spender'
    END AS spend_segment
  FROM customer_spend
)
SELECT
  spend_segment,
  COUNT(*) AS total_customers,
  ROUND(AVG(avg_spend), 2) AS avg_spend_per_segment
```

```
FROM segmented
GROUP BY spend_segment
ORDER BY avg_spend_per_segment DESC;
```

RESULT:

	spend_segment	total_customers ▾
1.	Medium Spender	45,167
2.	Low Spender	44,226
3.	High Spender	3,964



Insights:

- 48% of customers are Medium Spenders – stable and consistent.
- 47% are Low Spenders – high in volume, but low revenue impact.
- Only 4% are High Spenders, but likely contribute a large share of revenue.
- Focus on retaining High Spenders and upselling to Medium Spenders.

Summary of the Insights

1. Total Orders
    - **99441** total orders: confirms the data integrity across the orders table.
2. Top 10 Most Sold Products
    - aca2eb7d00ea1a7b8ebd4e68314663af has the highest sales count (527).
    - Great for product performance analysis — later you can join with the products table to get category or name info if available.
3. Total Revenue by Payment Type
    - Credit card dominates with over ₹12.5 million in revenue.
    - UPI follows, then vouchers and debit cards.
    - “not_defined” is likely bad/missing data — can be filtered or cleaned.
4. Best Reviewed Products
    - All listed products have perfect 5.0 ratings, but low review counts.
    - You can set a minimum review_count > 10 in your query to filter for consistently well-rated products.

Step 5: Add Product & Category Details to Insights

SQL:

```
SELECT
  oi.product_id,
  p.`product category` AS category,
  COUNT(*) AS order_count
FROM `scaler-bcstudy.ecommerce_analytics.order_item` oi
JOIN `scaler-bcstudy.ecommerce_analytics.products` p
  ON oi.product_id = p.product_id
GROUP BY oi.product_id, category
ORDER BY order_count DESC
LIMIT 10;
```

RESULTS:

product_id	category	order_count
aca2eb7d00ea1a7b8ebd4e68314663af	Furniture Decoration	527
99a4788cb24856965c36a24e339b6058	bed table bath	488
422879e10f46682990de24d770e7f83d	Garden tools	484
389d119b48cf3043d311335e499d9c6b	Garden tools	392
368c6c730842d78016ad823897a372db	Garden tools	388
53759a2ecddad2bb87a079a1f1519f73	Garden tools	373
d1c427060a0f73f6b889a5c7c61f2ac4	computer accessories	343
53b36df67ebb7c41585e8d54d6772e08	Watches present	323
154e7e31ebfa092203795c972e5804a6	HEALTH BEAUTY	281
3dd2a17168ec895c781a9191c1e95ad7	computer accessories	274

Steps 6: Time Series Analysis – Orders per Month

SQL:

```
SELECT
  FORMAT_DATE('%Y-%m', DATE(o.order_purchase_timestamp)) AS order_month,
  COUNT(*) AS total_orders
FROM `scaler-bcstudy.ecommerce_analytics.orders` o
GROUP BY order_month
ORDER BY order_month;
```

RESULTS:

order_month	total_orders
2016-09	4
2016-10	324

2016-12	1
2017-01	800
2017-02	1780
2017-03	2682
2017-04	2404
2017-05	3700
2017-06	3245
2017-07	4026
2017-08	4331
2017-09	4285
2017-10	4631
2017-11	7544
2017-12	5673
2018-01	7269
2018-02	6728
2018-03	7211
2018-04	6939
2018-05	6873
2018-06	6167
2018-07	6292
2018-08	6512
2018-09	16
2018-10	4

Step 7: Delivery Performance Analysis

SQL:

```
SELECT
  ROUND(AVG(DATE_DIFF(DATE(order_delivered_customer_date),
DATE(order_purchase_timestamp), DAY)), 2) AS avg_delivery_days
FROM `scaler-bcstudy.ecommerce_analytics.orders`
WHERE order_delivered_customer_date IS NOT NULL;
```

RESULTS:
avg_delivery_days
12.5

Step 8: Total Sales by Product Category

SQL:

```
SELECT
  p.`product category` AS category,
```

```
  COUNT(*) AS total_items_sold
FROM `scaler-bcstudy.ecommerce_analytics.order_item` oi
JOIN `scaler-bcstudy.ecommerce_analytics.products` p
  ON oi.product_id = p.product_id
GROUP BY category
ORDER BY total_items_sold DESC
LIMIT 10;
```

RESULTS:

category	total_items_sold
bed table bath	11115
HEALTH BEAUTY	9670
sport leisure	8641
Furniture Decoration	8334
computer accessories	7827
housewares	6964
Watches present	5991
telephony	4545
Garden tools	4347
automotive	4235

This Tells:

- Categories like bed table bath and health & beauty dominate your marketplace.
- Even though Garden tools had top-selling individual products, other categories beat it overall.

Step 9: Monthly Orders by Category

SQL:

```
SELECT
  FORMAT_DATE('%Y-%m', DATE(o.order_purchase_timestamp)) AS order_month,
  p.`product category` AS category,
  COUNT(*) AS items_sold
FROM `scaler-bcstudy.ecommerce_analytics.orders` o
JOIN `scaler-bcstudy.ecommerce_analytics.order_item` oi
  ON o.order_id = oi.order_id
JOIN `scaler-bcstudy.ecommerce_analytics.products` p
  ON oi.product_id = p.product_id
GROUP BY order_month, category
ORDER BY order_month, items_sold DESC;
```

RESULTS:

(A TABLE OF 1204 ROWS)

Step 10: Delivery Performance Analysis

Goal: Detect categories/sellers causing late deliveries

SQL:

```
SELECT
  p.`product category` AS category,
  ROUND(AVG(DATE_DIFF(DATE(o.order_delivered_customer_date),
DATE(o.order_purchase_timestamp), DAY)), 2) AS avg_delivery_days
FROM `scaler-bcstudy.ecommerce_analytics.orders` o
JOIN `scaler-bcstudy.ecommerce_analytics.order_item` oi
  ON o.order_id = oi.order_id
JOIN `scaler-bcstudy.ecommerce_analytics.products` p
  ON oi.product_id = p.product_id
WHERE o.order_delivered_customer_date IS NOT NULL
GROUP BY category
ORDER BY avg_delivery_days DESC
LIMIT 10;
```

RESULTS:

category	avg_delivery_days
Furniture office	20.79
Christmas articles	15.67
Fashion Calcados	15.4
insurance and services	15.0
House Comfort 2	14.53
CITTE AND UPHACK FURNITURE	14.41
ELECTRICES 2	13.86
Room Furniture	13.78
Garden tools	13.66
Fashion Underwear and Beach Fashion	13.61

Step 11: Revenue by Product Category

Goal: Prioritize high-value categories (not just high volume)

SQL:

```
SELECT
  p.`product category` AS category,
  ROUND(SUM(pay.payment_value), 2) AS total_revenue
FROM `scaler-bcstudy.ecommerce_analytics.payments` pay
JOIN `scaler-bcstudy.ecommerce_analytics.order_item` oi
  ON pay.order_id = oi.order_id
JOIN `scaler-bcstudy.ecommerce_analytics.products` p
  ON oi.product_id = p.product_id
GROUP BY category
ORDER BY total_revenue DESC
LIMIT 10;
```

RESULTS:

category	total_revenue
bed table bath	1712553.67
HEALTH BEAUTY	1657373.12
computer accessories	1585330.45
Furniture Decoration	1430176.39
Watches present	1429216.68
sport leisure	1392127.56
housewares	1094758.13
automotive	852294.33
Garden tools	838280.75
Cool Stuff	779698.0

Step 12: Repeat vs. One-Time Customers

Goal: Improve retention strategies

SQL:

```
SELECT
  COUNT(DISTINCT customer_id) AS total_customers,
  COUNT(*) AS total_orders,
  ROUND(COUNT(*) / COUNT(DISTINCT customer_id), 2) AS avg_orders_per_customer
FROM `scaler-bcstudy.ecommerce_analytics.orders`;
```

RESULTS:

total_customers	total_orders	avg_orders_per_customer
99441	99441	1.0

Step 13: Best/Worst Reviewed Categories

Goal: Understand customer satisfaction across product types

SQL:

```
SELECT
  p.`product category` AS category,
  ROUND(AVG(r.review_score), 2) AS avg_score,
  COUNT(*) AS total_reviews
FROM `scaler-bcstudy.ecommerce_analytics.order_reviews` r
JOIN `scaler-bcstudy.ecommerce_analytics.orders` o ON r.order_id = o.order_id
JOIN `scaler-bcstudy.ecommerce_analytics.order_item` oi ON o.order_id = oi.order_id
JOIN `scaler-bcstudy.ecommerce_analytics.products` p ON oi.product_id = p.product_id
GROUP BY category
HAVING COUNT(*) > 50
ORDER BY avg_score DESC;
```

RESULTS:

category	avg_score	total_reviews
General Interest Books	4.45	549
Construction Tools Tools	4.44	99
Imported books	4.4	60
technical books	4.37	266
Bags Accessories	4.32	1088
Drink foods	4.32	279
HOUSE PASTALS OVEN AND CAFE	4.3	76
Fashion Calcados	4.23	261
foods	4.22	495
cine photo	4.21	73
stationary store	4.19	2507
pet Shop	4.19	1939
home appliances	4.17	806
PCs	4.17	200
perfumery	4.16	3421
toys	4.16	4091
musical instruments	4.15	675
Cool Stuff	4.15	3772
electrostile	4.15	677
Fashion Bags and Accessories	4.14	2039
HEALTH BEAUTY	4.14	9645
ELECTRICES 2	4.14	238

Furniture	4.12	110
IMAGE IMPORT TABLETS	4.12	81
sport leisure	4.11	8640
Industry Commerce and Business	4.1	266
SIGNALIZATION AND SAFETY	4.09	197
Blu Ray DVDs	4.08	63
automotive	4.07	4213
housewares	4.06	6943
Construction Tools Illumination	4.05	296
Construction Tools Garden	4.05	240
drinks	4.05	377
Construction Tools Construction	4.05	926
Garden tools	4.04	4329
electronics	4.04	2749
Watches present	4.02	5950
Games consoles	4.02	1127
Market Place	4.02	309
Christmas articles	4.02	146
babies	4.01	3048
Agro Industria e Comercio	4.0	212
Fashion Underwear and Beach Fashion	3.98	130
climatization	3.97	292
Furniture Kitchen Service Area Dinner and Garden	3.96	280
telephony	3.95	4517
Art	3.94	207
Casa Construcao	3.94	600
computer accessories	3.93	7849
bed table bath	3.9	11137
Furniture Decoration	3.9	8331
Room Furniture	3.9	502
	3.84	1598
CONSTRUCTION SECURITY TOOLS	3.84	193
audio	3.83	361
House comfort	3.83	435
fixed telephony	3.68	262
Fashion Men's Clothing	3.64	131
Furniture office	3.49	1687

Summary of Business Findings

1. Delivery Performance Analysis

Category	Avg Delivery Days
Furniture office	20.79 *(slowest)*
Christmas articles	15.67
Fashion Calcados	15.40
Room Furniture	13.78
Garden tools	13.66

Insight: Delivery delays are significant in large or seasonal items. Suggest:

- Buffering lead times for furniture/fashion
- Partnering with logistics vendors for faster delivery

2. Revenue by Product Category

Category	Revenue
Bed table bath	₹1.71M
Health & Beauty	₹1.65M
Computer Accessories	₹1.58M
Furniture Decoration	₹1.43M
Watches present	₹1.42M

Insight: Focus on high-revenue but lower-rated categories (see below) to improve UX and retention.

3. Repeat vs One-Time Customers

Metric	Value
Total Customers	99,441
Avg Orders/Customer	1.0

Critical Business Insight: Nearly all are **one-time buyers**.
Suggestion:

- Loyalty programs
- Email/WhatsApp re-engagement
- Personalized recommendations to boost retention

4. Customer Satisfaction by Category (Avg Ratings)

Top Rated	Low Rated
General Interest Books (4.45)	Furniture Office (3.49)
Imported books (4.40)	Fashion Men's Clothing (3.64)
Perfumery (4.16)	Fixed telephony (3.68)
Health & Beauty (4.14)	Furniture (3.90)
Sport Leisure (4.11)	Bed table bath (3.90)

Insight:

- Low ratings in revenue-heavy categories (e.g., Bed Table Bath, Furniture Office) indicate poor post-purchase experience.
- Recommend better product curation, reviews moderation, or UX feedback surveys.

Final Recommendations for Your Business Study:

Area	Recommendation
Marketing	Promote top-rated, high-growth categories (books, perfumery)
Logistics	Optimize delivery for bulky/seasonal products
Product Strategy	Improve support and QC for low-rated but high-selling items
Retention	Launch loyalty programs and customer re-engagement
Growth Tracking	Use monthly time-series dashboards for performance forecasting

Step 14: SQL for Time of Day Order Pattern

SQL:

```
SELECT
  CASE
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 0 AND 6 THEN
'Dawn'
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 7 AND 12
THEN 'Morning'
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 13 AND 18
THEN 'Afternoon'
    WHEN EXTRACT(HOUR FROM order_purchase_timestamp) BETWEEN 19 AND 23
THEN 'Night'
  END AS time_of_day,
  COUNT(*) AS total_orders
FROM `scaler-bcstudy.ecommerce_analytics.orders`
WHERE order_status = 'delivered' -- only considering completed orders
GROUP BY time_of_day
ORDER BY total_orders DESC;
```

RESULTS:

time_of_day	total_orders
Afternoon	36965
Night	27522
Morning	26919
Dawn	5072

Insight:
Most Brazilian customers prefer to shop **in the afternoon**, likely after work or school hours. Very few orders are placed during the dawn period.

Step 15: SQL: Cost Increase from Jan–Aug 2017 to Jan–Aug 2018

SQL:

```
WITH payments_with_year_month AS (
  SELECT
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year,
    EXTRACT(MONTH FROM o.order_purchase_timestamp) AS month,
    p.payment_value
  FROM `scaler-bcstudy.ecommerce_analytics.orders` o
  JOIN `scaler-bcstudy.ecommerce_analytics.payments` p
    ON o.order_id = p.order_id
  WHERE o.order_status = 'delivered'
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND
8
```

```
    AND EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
)

SELECT
  year,
  ROUND(SUM(payment_value), 2) AS total_payment_value
FROM payments_with_year_month
GROUP BY year
ORDER BY year;
```

RESULTS:

year	total_payment_value
2017	3473862.76
2018	8452975.2

Insight: Year-on-Year Payment Growth (Jan–Aug)

Total payment value (delivered orders only):

- **2017 (Jan–Aug):** R$ 3.47 million
- **2018 (Jan–Aug):** R$ 8.45 million

Growth: ~143.3% increase in total revenue — a strong indicator of e-commerce expansion in Brazil.

Step 16: Calculate % Increase:

```
WITH payments_summary AS (
  -- Same CTE as before
  SELECT
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS year,
    p.payment_value
  FROM `scaler-bcstudy.ecommerce_analytics.orders` o
  JOIN `scaler-bcstudy.ecommerce_analytics.payments` p
    ON o.order_id = p.order_id
  WHERE o.order_status = 'delivered'
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND
8
    AND EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
),
agg AS (
  SELECT
    year,
```

```
    ROUND(SUM(payment_value), 2) AS total_payment
  FROM payments_summary
  GROUP BY year
)
SELECT
  MAX(CASE WHEN year = 2017 THEN total_payment END) AS total_2017,
  MAX(CASE WHEN year = 2018 THEN total_payment END) AS total_2018,
  ROUND(((MAX(CASE WHEN year = 2018 THEN total_payment END) -
        MAX(CASE WHEN year = 2017 THEN total_payment END)) /
        MAX(CASE WHEN year = 2017 THEN total_payment END)) * 100, 2) AS
percentage_increase
FROM agg;
```

RESULTS:

total_2017	total_2018	percentage_increase
3473862.76	8452975.2	143.33

"Impact on Economy" — % Increase in Order Costs (2017 to 2018)

Insight: Year-on-Year Cost Growth (Jan–Aug)

- 2017 Total Payment Value: R$ 3,473,862.76
- 2018 Total Payment Value: R$ 8,452,975.20
- Percentage Increase: 143.33%

Step 17: SQL: Orders by Payment Type Per Month

SQL:

```
SELECT
  FORMAT_DATE('%Y-%m', DATE(o.order_purchase_timestamp)) AS order_month,
  p.payment_type,
  COUNT(DISTINCT o.order_id) AS total_orders
FROM `scaler-bcstudy.ecommerce_analytics.orders` o
JOIN `scaler-bcstudy.ecommerce_analytics.payments` p
  ON o.order_id = p.order_id
WHERE o.order_status = 'delivered'
GROUP BY order_month, p.payment_type
ORDER BY order_month, total_orders DESC;
```

RESULTS:

order_month	payment_type	total_orders
2016-10	credit_card	208
2016-10	UPI	51
2016-10	voucher	9
2016-10	debit_card	2
2016-12	credit_card	1
2017-01	credit_card	541
2017-01	UPI	188
2017-01	voucher	32
2017-01	debit_card	9
2017-02	credit_card	1249
2017-02	UPI	371
2017-02	voucher	63
2017-02	debit_card	13
2017-03	credit_card	1901
2017-03	UPI	565
2017-03	voucher	120
2017-03	debit_card	30
2017-04	credit_card	1761
2017-04	UPI	474
2017-04	voucher	107
2017-04	debit_card	25
2017-05	credit_card	2714
2017-05	UPI	740
2017-05	voucher	168
2017-05	debit_card	29
2017-06	credit_card	2362
2017-06	UPI	689
2017-06	voucher	140
2017-06	debit_card	26
2017-07	credit_card	2960
2017-07	UPI	811
2017-07	voucher	193
2017-07	debit_card	20
2017-08	credit_card	3174
2017-08	UPI	902
2017-08	voucher	189
2017-08	debit_card	33
2017-09	credit_card	3175
2017-09	UPI	868

2017-09	voucher	171
2017-09	debit_card	43
2017-10	credit_card	3402
2017-10	UPI	955
2017-10	voucher	196
2017-10	debit_card	51
2017-11	credit_card	5686
2017-11	UPI	1445
2017-11	voucher	257
2017-11	debit_card	65
2017-12	credit_card	4232
2017-12	UPI	1134
2017-12	voucher	214
2017-12	debit_card	62
2018-01	credit_card	5360
2018-01	UPI	1473
2018-01	voucher	292
2018-01	debit_card	109
2018-02	credit_card	5096
2018-02	UPI	1294
2018-02	voucher	214
2018-02	debit_card	68
2018-03	credit_card	5509
2018-03	UPI	1316
2018-03	voucher	264
2018-03	debit_card	74
2018-04	credit_card	5327
2018-04	UPI	1265
2018-04	voucher	235
2018-04	debit_card	94
2018-05	credit_card	5376
2018-05	UPI	1242
2018-05	voucher	199
2018-05	debit_card	49
2018-06	credit_card	4743
2018-06	UPI	1089
2018-06	voucher	228
2018-06	debit_card	179
2018-07	credit_card	4644
2018-07	UPI	1200

2018-07	debit_card	234
2018-07	voucher	209
2018-08	credit_card	4883
2018-08	UPI	1119
2018-08	debit_card	270
2018-08	voucher	179

Insight: Payment Methods Over Time

- Credit Card is by far the most used payment method every month.
- UPI (boleto) is the second most common, gradually increasing.
- Vouchers and debit cards are used less frequently.
- The number of orders rises significantly from late 2017 onward — supporting the growth trend.

STEP 18: SQL: Orders by Installment Count

SQL:

```
SELECT
  payment_installments,
  COUNT(DISTINCT order_id) AS total_orders
FROM `scaler-bcstudy.ecommerce_analytics.payments`
GROUP BY payment_installments
ORDER BY payment_installments;
```

RESULTS:

Insight: Installment-Based Orders

Installments	Total Orders
1 installment	**49,060** orders
2–6 installments	Moderate volume
10 installments	5,315 orders
12+ installments	Rare (under 150 total)
0 installments	Only 2 cases (possibly test or error data)

Interpretation:
The majority of customers prefer to pay in full or with short-term installments (1–3). Long-term installments (12–24 months) are rarely used.

Step 19: Conclusion & Recommendations:

Based on the data from 100,000 orders across Brazil:

- Afternoon is the most popular time to shop online
- Credit card is the dominant payment method
- Top categories like “bed table bath” and “HEALTH BEAUTY” drive the most revenue
- SP and RJ are the highest contributing states
- Delivery delays and high freight costs in some states can be improved
- 1-time customers dominate — loyalty programs can help increase retention

Target can optimize operations by focusing on:

- Regional delivery improvements
- High-demand categories
- Customer loyalty strategies